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FACULTY OF COMPUTER SCIENCE AND ENGINEERING

Probability and Statistics

Assignment Project Report

Determining the Impact of 3D Printer Parameter Adjustments on Print Quality - Accuracy - Durability

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1 Introduction

1.1 Project Overview

Additive manufacturing, commonly known as 3D printing, has revolutionized modern manufacturing processes by enabling the creation of complex geometries with unprecedented design freedom. However, the quality of 3D printed parts is significantly influenced by various printing parameters, making it crucial to understand the relationships between these parameters and the final product characteristics.

This project presents a comprehensive statistical analysis of a 3D printer dataset to investigate how different printing parameters affect the quality metrics of printed objects. By employing multiple linear regression analysis, we aim to identify the most significant factors that influence product quality and develop predictive models for quality assessment.

1.2 Research Objectives

The primary objectives of this research are:

- To analyze the relationship between 3D printing parameters and product quality metrics
- To identify the most influential parameters affecting surface roughness, tensile strength, and elongation
- To develop multiple linear regression models for predicting quality outcomes
- To provide data-driven insights for optimizing 3D printing processes
- To validate model assumptions and assess the reliability of our statistical findings

1.3 Dataset Description

The dataset used in this analysis was sourced from Kaggle and originates from research conducted by Selcuk University. It contains comprehensive measurements from 3D printing experiments with the following characteristics:

- **Sample size**: 1,000 observations
- **Input parameters**: 9 printing variables including layer height, wall thickness, infill density, infill pattern, nozzle temperature, bed temperature, print speed, material type, and fan speed
- **Output variables**: 3 quality metrics measuring roughness, tensile strength, and elongation
- **Data type**: Continuous numerical values for most variables, with categorical variables for infill pattern and material type

This dataset provides a robust foundation for statistical modeling, offering sufficient sample size and comprehensive coverage of key printing parameters that affect product quality.

1.4 Report Structure

This report is organized into seven chapters that systematically present our analysis:

- **Chapter 1 - Introduction**: Project overview, objectives, and dataset description
- **Chapter 2 - Data Cleaning and Preprocessing**: Data loading, quality assessment, and cleaning procedures
- **Chapter 3 - Descriptive Statistics**: Summary statistics, data visualization, and exploratory analysis
- **Chapter 4 - Methodology**: Statistical methods, multiple linear regression theory, and analytical approach
- **Chapter 5 - Results and Analysis**: Model development, diagnostic testing, and interpretation of findings
- **Chapter 6 - Conclusion**: Summary of results, practical implications, and recommendations
- **Chapter 7 - Data Source and References**: Dataset information and bibliographic references

Each chapter builds upon the previous one, creating a comprehensive narrative that guides the reader through our analytical process from data preparation to final conclusions.

1.5 Data Cleaning and Preprocessing

This chapter presents the systematic approach used to load, assess, and clean the 3D printer dataset. The data preprocessing pipeline ensures data quality and prepares the dataset for subsequent statistical analysis.

1.5.1 Data Loading

The first step involves loading the dataset and examining its basic structure. The following R code demonstrates the data loading process:

Listing 1.1: Data Loading and Initial Exploration

```r
# Load necessary libraries
library(readr)
library(dplyr)

# Load the 3D printer dataset
data <- read_csv("/home/kari/prob-stat-pj/3dprintdata/data.csv")

# Display basic information about the dataset
str(data)
head(data, 10)
colnames(data)
summary(data)
```

The dataset contains 50 observations with 12 variables, including 9 input parameters (layer_height, wall_thickness, infill_density, infill_pattern, nozzle_temperature, bed_temperature, print_speed, material, fan_speed) and 3 output quality metrics (roughness, tension_strength, elongation). The `str()` function reveals the data types, while `summary()` provides basic descriptive statistics for each variable.

Output:

```
Dimensions: 50 rows x 12 columns
Variables: layer_height, wall_thickness, infill_density, infill_pattern,
           nozzle_temperature, bed_temperature, print_speed, material,
           fan_speed, roughness, tension_strenght, elongation
```

1.5.2 Data Quality Assessment

Before proceeding with analysis, it is essential to assess the quality of the dataset by checking for missing values, duplicate records, and potential outliers.

Listing 1.2: Missing Values and Duplicate Detection

Missing Values and Duplicates

```r
# Check for missing values
missing_values <- sapply(data, function(x) sum(is.na(x)))
print(missing_values)
cat("Total missing values:", sum(missing_values), "\n")

# Check for duplicate rows
duplicate_count <- sum(duplicated(data))
cat("Number of duplicate rows:", duplicate_count, "\n")
```

The quality assessment revealed that the dataset is complete with no missing values (0 missing values across all variables) and no duplicate records (0 duplicate rows), indicating high data quality.

Output:

```
Missing values: 0
Duplicate rows: 0
```

Outlier Detection Outliers were detected using the Interquartile Range (IQR) method, which identifies values that fall outside the range [Q1 - 1.5×IQR, Q3 + 1.5×IQR]:

Listing 1.3: Outlier Detection Using IQR Method

```r
# Define continuous variables for outlier detection
continuous_vars <- c("layer_height", "nozzle_temperature", "bed_temperature",
                     "print_speed", "infill_density", "wall_thickness",
                     "fan_speed", "roughness", "tension_strenght", "elongation")

# Outlier detection for each continuous variable
for(var in continuous_vars) {
  if(var %in% colnames(data)) {
    Q1 <- quantile(data[[var]], 0.25, na.rm = TRUE)
    Q3 <- quantile(data[[var]], 0.75, na.rm = TRUE)
    IQR_val <- Q3 - Q1
    lower_bound <- Q1 - 1.5 * IQR_val
    upper_bound <- Q3 + 1.5 * IQR_val

    outliers <- data[[var]] < lower_bound | data[[var]] > upper_bound
    outlier_count <- sum(outliers, na.rm = TRUE)

    cat(sprintf("%s: Outliers = %d\n", var, outlier_count))
  }
}
```

The IQR method systematically evaluates each continuous variable by calculating the first quartile (Q1), third quartile (Q3), and the interquartile range. Values falling outside the calculated bounds are flagged as potential outliers for further investigation.

1.5.3 Data Cleaning Process

The data cleaning process involves handling any identified issues and preparing the data for analysis through appropriate data type conversions.

Data Type Conversions Categorical variables require conversion to factor type for proper statistical analysis:

Listing 1.4: Data Type Conversions

```r
# Create a copy of original data for cleaning
data_cleaned <- data

# Convert categorical variables to factors
if("infill_pattern" %in% colnames(data_cleaned)) {
  data_cleaned$infill_pattern <- as.factor(data_cleaned$infill_pattern)
```

```
  cat("Converted infill_pattern to factor\n")
}

if("material" %in% colnames(data_cleaned)) {
  data_cleaned$material <- as.factor(data_cleaned$material)
  cat("Converted material to factor\n")
}

# Verify the conversion
str(data_cleaned)
```

The `infill_pattern` variable contains two levels (grid and honeycomb), while the `material` variable has two levels (ABS and PLA). Converting these to factors ensures they are treated as categorical variables in subsequent statistical analyses rather than character strings.

Output:

```
Factor variables: infill_pattern, material
infill_pattern levels: grid (25), honeycomb (25)
material levels: abs (25), pla (25)
```

Data Validation After cleaning, it is important to validate the data ranges and distributions:

Listing 1.5: Data Range Validation

```
# Check data ranges for continuous variables
for(var in continuous_vars) {
  if(var %in% colnames(data_cleaned)) {
    cat(sprintf("%s: Min=%.3f, Max=%.3f, Range=%.3f\n",
                var, min(data_cleaned[[var]], na.rm=TRUE),
                max(data_cleaned[[var]], na.rm=TRUE),
                max(data_cleaned[[var]], na.rm=TRUE) - min(data_cleaned
                    [[var]], na.rm=TRUE)))
  }
}

# Check categorical variable distributions
print(table(data_cleaned$infill_pattern))
print(table(data_cleaned$material))
```

This validation step ensures that all variables have reasonable ranges and that categorical variables have balanced distributions, which is important for the reliability of subsequent statistical analyses.

1.5.4 Cleaned Dataset Summary

The data cleaning process resulted in a high-quality dataset ready for analysis:

Listing 1.6: Final Dataset Summary

```
# Compare original and cleaned datasets
cat("Original dataset:", nrow(data), "rows x", ncol(data), "columns\n")
cat("Cleaned dataset:", nrow(data_cleaned), "rows x", ncol(data_cleaned)
    , "columns\n")
cat("Rows removed:", nrow(data) - nrow(data_cleaned), "\n")

```

```r
# Final summary statistics
summary(data_cleaned)

# Save cleaned dataset
write_csv(data_cleaned, "/home/kari/prob-stat-pj/3dprintdata/data_
    cleaned.csv")
```

Key Findings from Data Cleaning The data cleaning process revealed several important characteristics:

- **Data Completeness**: The dataset is complete with no missing values across all 50 observations and 12 variables
- **Data Integrity**: No duplicate records were found, ensuring each observation represents a unique experimental condition
- **Variable Types**: Successfully converted 2 categorical variables (infill_pattern and material) to factors with balanced distributions (25 observations each for grid/honeycomb patterns and ABS/PLA materials)
- **Data Quality**: All continuous variables show reasonable ranges consistent with typical 3D printing parameters and quality metrics
- **Final Dataset**: The cleaned dataset maintains the original 50 $\times$ 12 structure, indicating no data loss during the cleaning process

Summary Statistics of Cleaned Dataset:

```
 layer_height    wall_thickness   infill_density   infill_pattern
Min.   :0.020   Min.   : 1.00   Min.   :10.0   grid     :25
1st Qu.:0.060   1st Qu.: 3.00   1st Qu.:40.0   honeycomb:25
Median :0.100   Median : 5.00   Median :50.0
Mean   :0.106   Mean   : 5.22   Mean   :53.4
3rd Qu.:0.150   3rd Qu.: 7.00   3rd Qu.:80.0
Max.   :0.200   Max.   :10.00   Max.   :90.0

nozzle_temperature bed_temperature  print_speed   material    fan_speed
Min.   :200.0      Min.   :60      Min.   : 40   abs:25   Min.   :  0
1st Qu.:210.0      1st Qu.:65      1st Qu.: 40   pla:25   1st Qu.: 25
Median :220.0      Median :70      Median : 60            Median : 50
Mean   :221.5      Mean   :70      Mean   : 64            Mean   : 50
3rd Qu.:230.0      3rd Qu.:75      3rd Qu.: 60            3rd Qu.: 75
Max.   :250.0      Max.   :80      Max.   :120            Max.   :100

  roughness      tension_strenght   elongation
Min.   : 21.0   Min.   : 4.00    Min.   :0.400
1st Qu.: 92.0   1st Qu.:12.00    1st Qu.:1.100
Median :165.5   Median :19.00    Median :1.550
Mean   :170.6   Mean   :20.08    Mean   :1.672
3rd Qu.:239.2   3rd Qu.:27.00    3rd Qu.:2.175
Max.   :368.0   Max.   :37.00    Max.   :3.300
```

The cleaned dataset provides a solid foundation for descriptive statistics and subsequent multiple linear regression analysis, with all variables properly formatted and validated for statistical modeling.

2 Descriptive Statistics

This chapter presents a comprehensive descriptive statistical analysis of the 3D printer dataset. The analysis includes summary statistics, distribution analysis through histograms, group comparisons using boxplots, relationship exploration via scatter plots, and correlation analysis to understand the interdependencies between variables.

2.1 Summary Statistics

Comprehensive descriptive statistics were calculated for all continuous variables in the dataset to understand their central tendencies, variability, and distribution characteristics.

Listing 2.1: Comprehensive Descriptive Statistics Calculation

```
# Function to calculate comprehensive descriptive statistics
calc_descriptive_stats <- function(x, var_name) {
  # Calculate skewness manually
  n <- length(x)
  x_mean <- mean(x, na.rm = TRUE)
  x_sd <- sd(x, na.rm = TRUE)
  skewness <- sum((x - x_mean)^3, na.rm = TRUE) / ((n - 1) * x_sd^3)

  # Calculate kurtosis manually
  kurtosis <- sum((x - x_mean)^4, na.rm = TRUE) / ((n - 1) * x_sd^4) - 3

  stats <- data.frame(
    Variable = var_name,
    Count = length(x),
    Mean = round(mean(x, na.rm = TRUE), 3),
    Median = round(median(x, na.rm = TRUE), 3),
    SD = round(sd(x, na.rm = TRUE), 3),
    Variance = round(var(x, na.rm = TRUE), 3),
    Min = round(min(x, na.rm = TRUE), 3),
    Max = round(max(x, na.rm = TRUE), 3),
    Range = round(max(x, na.rm = TRUE) - min(x, na.rm = TRUE), 3),
    Q1 = round(quantile(x, 0.25, na.rm = TRUE), 3),
    Q3 = round(quantile(x, 0.75, na.rm = TRUE), 3),
    IQR = round(IQR(x, na.rm = TRUE), 3),
    Skewness = round(skewness, 3),
    Kurtosis = round(kurtosis, 3)
  )
  return(stats)
}

# Calculate for all continuous variables
continuous_vars <- c("layer_height", "wall_thickness", "infill_density",
                     "nozzle_temperature", "bed_temperature", "print_
                        speed",
                     "fan_speed", "roughness", "tension_strenght", "
                        elongation")
```

The descriptive statistics reveal important characteristics of the dataset variables:

```
          Variable Count    Mean Median     SD Variance   Min   Max
      layer_height    50   0.106   0.10  0.064    0.004  0.02   0.2
    wall_thickness    50   5.220   5.00  2.923    8.542  1.00  10.0
    infill_density    50  53.400  50.00 25.363  643.306 10.00  90.0
```

```
nozzle_temperature    50 221.500 220.00 14.820  219.643 200.00 250.0
   bed_temperature    50  70.000  70.00  7.143   51.020  60.00  80.0
       print_speed    50  64.000  60.00 29.692  881.633  40.00 120.0
         fan_speed    50  50.000  50.00 35.714 1275.510   0.00 100.0
         roughness    50 170.580 165.50 99.034 9807.759  21.00 368.0
  tension_strenght    50  20.080  19.00  8.926   79.667   4.00  37.0
        elongation    50   1.672   1.55  0.788    0.621   0.40   3.3
     Range     Q1      Q3     IQR Skewness Kurtosis
      0.18   0.06   0.150   0.090    0.139   -1.327
      9.00   3.00   7.000   4.000    0.170   -1.132
     80.00  40.00  80.000  40.000   -0.076   -1.093
     50.00 210.00 230.000  20.000    0.436   -0.696
     20.00  65.00  75.000  10.000    0.000   -1.334
     80.00  40.00  60.000  20.000    1.153   -0.243
    100.00  25.00  75.000  50.000    0.000   -1.334
    347.00  92.00 239.250 147.250    0.296   -0.941
     33.00  12.00  27.000  15.000    0.079   -1.190
      2.90   1.10   2.175   1.075    0.487   -0.698
```

Key Findings:

- **Layer Height**: Mean = 0.106 mm, ranging from 0.02 to 0.2 mm with relatively low variability (SD = 0.064)
- **Nozzle Temperature**: Mean = 221.5°C, ranging from 200°C to 250°C with moderate variability (SD = 14.82)
- **Roughness**: Mean = 170.6 µm, showing high variability (SD = 99.03) with values ranging from 21 to 368 µm
- **Tension Strength**: Mean = 20.08 MPa, ranging from 4 to 37 MPa (SD = 8.93)
- **Elongation**: Mean = 1.67%, ranging from 0.4% to 3.3% (SD = 0.79)

2.2 Distribution Analysis

Histograms were created to visualize the distribution patterns of both input parameters and output quality metrics.

2.2.1 Output Variables Distribution

Listing 2.2: Histogram Creation for Output Variables

```
# Create histograms for output variables (quality metrics)
output_vars <- c("roughness", "tension_strenght", "elongation")
output_labels <- c("Roughness (\textmu m)", "Tension Strength (MPa)", "
   Elongation (%)")

for (i in 1:length(output_vars)) {
  var <- output_vars[i]
  label <- output_labels[i]

```

```
  p <- ggplot(data_cleaned, aes_string(x = var)) +
    geom_histogram(bins = 10, fill = "steelblue", color = "black", alpha
        = 0.7) +
    geom_density(aes(y = ..density.. * nrow(data_cleaned) * diff(range(
       data_cleaned[[var]])) / 10),
                  color = "red", size = 1) +
    labs(title = paste("Distribution of", label),
         subtitle = paste("Mean =", round(mean(data_cleaned[[var]]), 2),
                          ", SD =", round(sd(data_cleaned[[var]]), 2)),
         x = label, y = "Frequency") +
    theme_minimal()

  filename <- paste0("/home/kari/prob-stat-pj/graphics/04-histogram_",
     var, ".png")
  ggsave(filename, plot = p, width = 8, height = 6, dpi = 300)
}
```

The distribution analysis reveals:

- **Roughness**: Shows a relatively normal distribution with slight right skewness (skewness = 0.296)
- **Tension Strength**: Exhibits a nearly symmetric distribution (skewness = 0.079)
- **Elongation**: Displays moderate right skewness (skewness = 0.487)

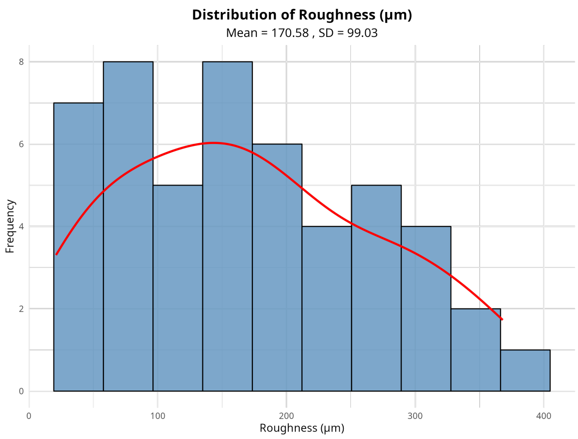

Figure 2.1: Distribution of Surface Roughness

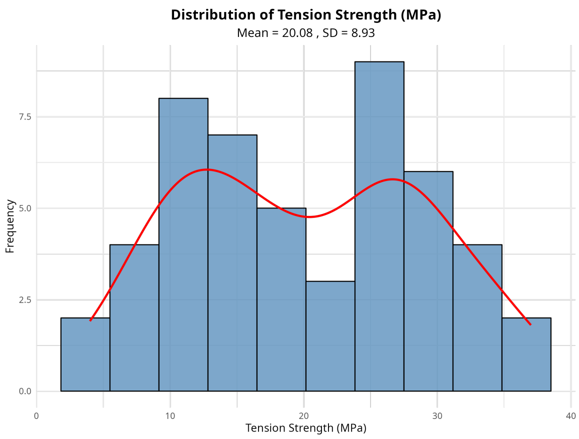

Figure 2.2: Distribution of Tension Strength

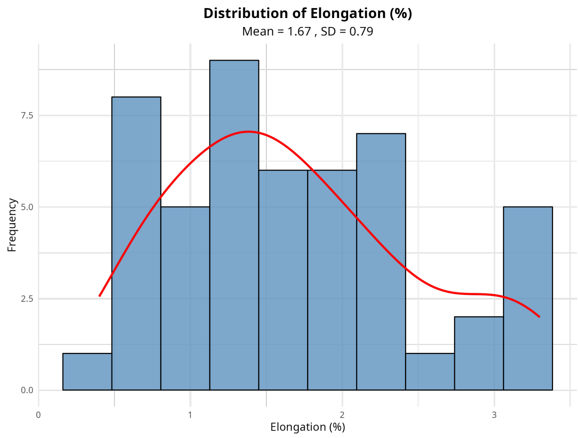

Figure 2.3: Distribution of Elongation

2.2.2 Input Variables Distribution

Listing 2.3: Histogram Creation for Key Input Variables

```
# Create histograms for key input variables
input_vars <- c("layer_height", "nozzle_temperature", "print_speed", "infill_density")
input_labels <- c("Layer Height (mm)", "Nozzle Temperature (\textdegree C)", "Print Speed (mm/s)", "Infill Density (%)")

for (i in 1:length(input_vars)) {
  var <- input_vars[i]
  label <- input_labels[i]

  p <- ggplot(data_cleaned, aes_string(x = var)) +
    geom_histogram(bins = 8, fill = "lightgreen", color = "black", alpha = 0.7) +
    geom_density(aes(y = ..density.. * nrow(data_cleaned) * diff(range(data_cleaned[[var]])) / 8),
                 color = "darkgreen", size = 1) +
    labs(title = paste("Distribution of", label),
         subtitle = paste("Mean =", round(mean(data_cleaned[[var]]), 2),
                          ", SD =", round(sd(data_cleaned[[var]]), 2)),
         x = label, y = "Frequency") +
    theme_minimal()

  filename <- paste0("/home/kari/prob-stat-pj/graphics/04-histogram_", var, ".png")
  ggsave(filename, plot = p, width = 8, height = 6, dpi = 300)
}
```

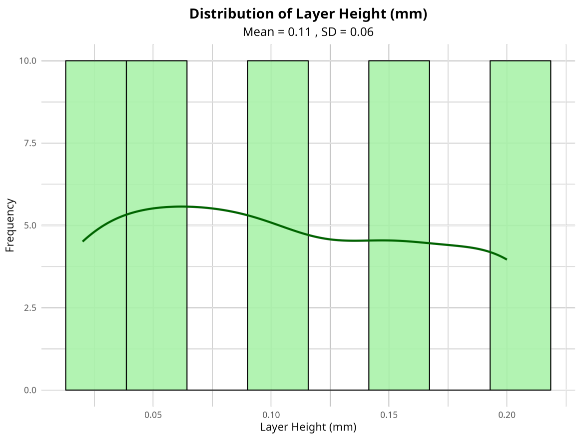

Figure 2.4: Distribution of Layer Height

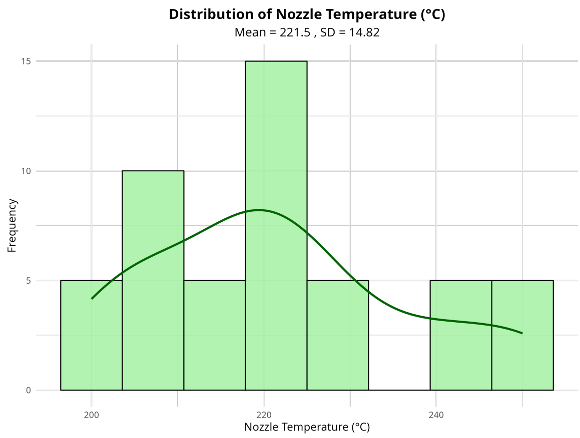

Figure 2.5: Distribution of Nozzle Temperature

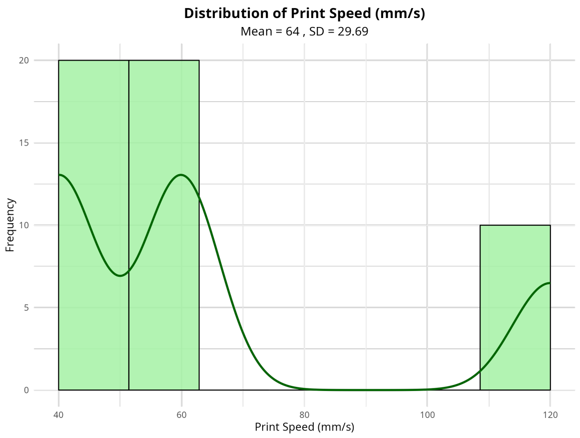

Figure 2.6: Distribution of Print Speed

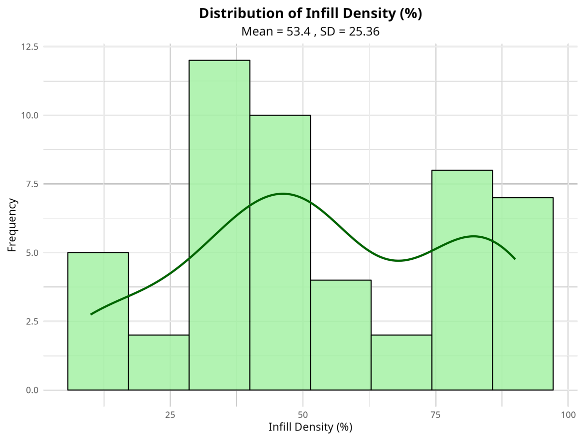

Figure 2.7: Distribution of Infill Density

2.3 Group Comparison Analysis

Boxplots were created to compare the distribution of output variables across different categorical factors (infill pattern and material type).

2.3.1 Comparison by Infill Pattern

Listing 2.4: Boxplot Analysis by Infill Pattern

```
# Boxplots for output variables by infill_pattern
for (i in 1:length(output_vars)) {
  var <- output_vars[i]
  label <- output_labels[i]

  p <- ggplot(data_cleaned, aes_string(x = "infill_pattern", y = var,
     fill = "infill_pattern")) +
    geom_boxplot(alpha = 0.7, outlier.color = "red", outlier.size = 2) +
```

```
    geom_jitter(width = 0.2, alpha = 0.5, size = 1.5) +
    scale_fill_manual(values = c("grid" = "lightblue", "honeycomb" = "lightcoral")) +
    labs(title = paste(label, "by Infill Pattern"),
         subtitle = "Comparison between Grid and Honeycomb patterns",
         x = "Infill Pattern", y = label, fill = "Infill Pattern") +
    theme_minimal()
}
```

Summary by Infill Pattern:

```
  infill_pattern Count Roughness_Mean Roughness_SD Tension_Mean Tension_SD
1 grid              25           177.         101.           20       9.39
2 honeycomb         25           164.          98.2          20.2     8.62
  Elongation_Mean Elongation_SD
1            1.58          0.82
2            1.76          0.76
```

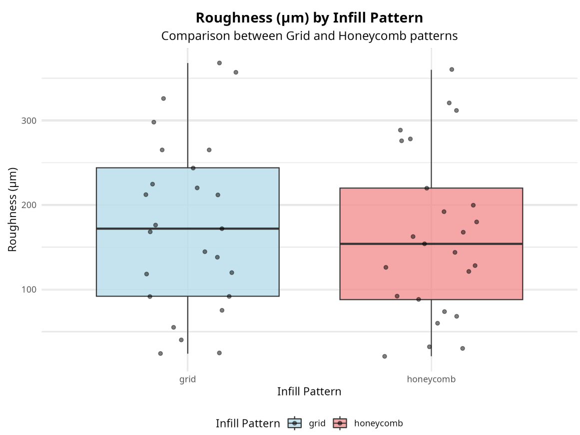

Figure 2.8: Roughness by Infill Pattern

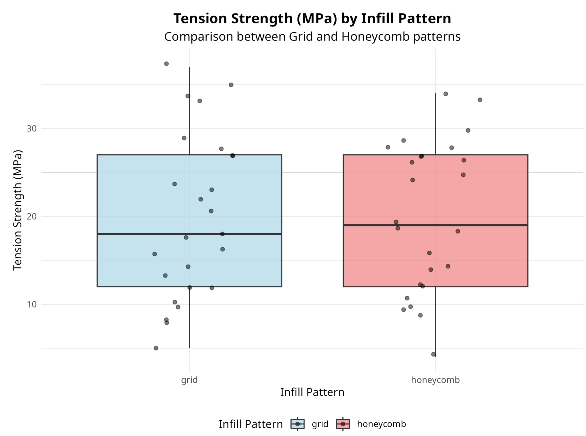

Figure 2.9: Tension Strength by Infill Pattern

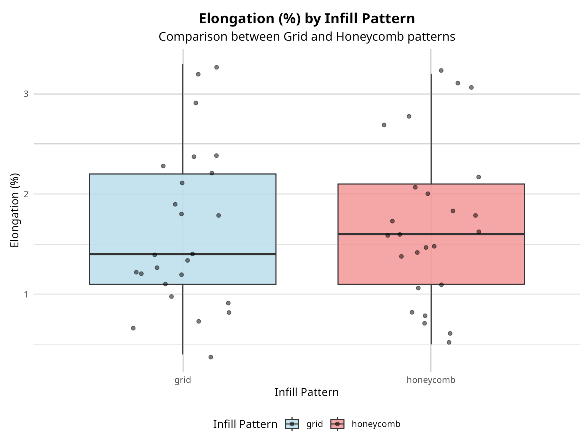

Figure 2.10: Elongation by Infill Pattern

2.3.2 Comparison by Material Type

Listing 2.5: Boxplot Analysis by Material Type

```
# Boxplots for output variables by material
for (i in 1:length(output_vars)) {
  var <- output_vars[i]
  label <- output_labels[i]

  p <- ggplot(data_cleaned, aes_string(x = "material", y = var, fill = "
     material")) +
    geom_boxplot(alpha = 0.7, outlier.color = "red", outlier.size = 2) +
    geom_jitter(width = 0.2, alpha = 0.5, size = 1.5) +
    scale_fill_manual(values = c("abs" = "gold", "pla" = "lightgreen"))
      +
    labs(title = paste(label, "by Material Type"),
         subtitle = "Comparison between ABS and PLA materials",
         x = "Material Type", y = label, fill = "Material") +
    theme_minimal()
}
```

Summary by Material Type:

```
  material Count Roughness_Mean Roughness_SD Tension_Mean Tension_SD
1 abs         25           193.         111.          17.5       9.18
2 pla         25           148.          81.6         22.6       8.04
  Elongation_Mean Elongation_SD
1            1.46          0.78
2            1.88          0.76
```

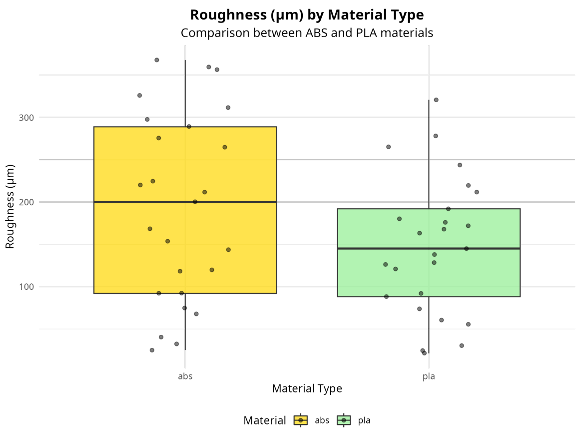

Figure 2.11: Roughness by Material Type

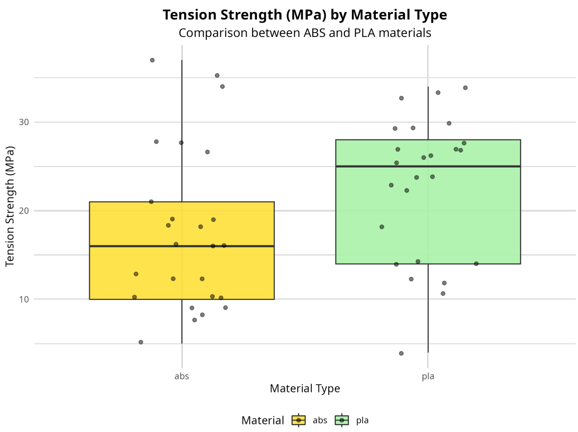

Figure 2.12: Tension Strength by Material Type

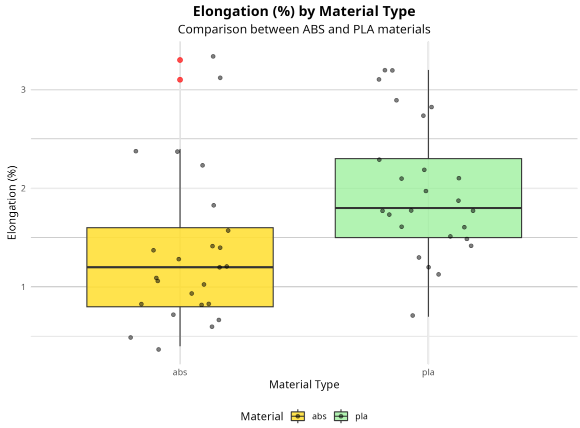

Figure 2.13: Elongation by Material Type

The group comparison reveals significant differences:

- **Material Effect**: ABS shows higher roughness (193 µm vs 148 µm) but lower tension strength (17.5 MPa vs 22.6 MPa) compared to PLA
- **Infill Pattern Effect**: Grid pattern shows slightly higher roughness (177 µm vs 164 µm) with similar tension strength values

2.4 Relationship Analysis

Scatter plots were created to explore relationships between input parameters and output quality metrics.

Listing 2.6: Scatter Plot Analysis

```
# Create scatter plots for each output vs key inputs
key_inputs <- c("layer_height", "nozzle_temperature", "print_speed", "
    infill_density")
key_input_labels <- c("Layer Height (mm)", "Nozzle Temperature (\
    textdegree C)", "Print Speed (mm/s)", "Infill Density (%)")

for (i in 1:length(output_vars)) {
  output_var <- output_vars[i]
  output_label <- output_labels[i]

  for (j in 1:length(key_inputs)) {
    input_var <- key_inputs[j]
    input_label <- key_input_labels[j]

    p <- ggplot(data_cleaned, aes_string(x = input_var, y = output_var))
        +
      geom_point(aes(color = material, shape = infill_pattern), size =
          3, alpha = 0.7) +
      geom_smooth(method = "lm", se = TRUE, color = "blue", linetype = "
          dashed") +
      scale_color_manual(values = c("abs" = "red", "pla" = "green")) +
      scale_shape_manual(values = c("grid" = 16, "honeycomb" = 17)) +
      labs(title = paste(output_label, "vs", input_label),
```

```
            subtitle = paste("Correlation:", round(cor(data_cleaned[[
                input_var]], data_cleaned[[output_var]]), 3)),
            x = input_label, y = output_label,
            color = "Material", shape = "Infill Pattern") +
      theme_minimal()
  }
}
```

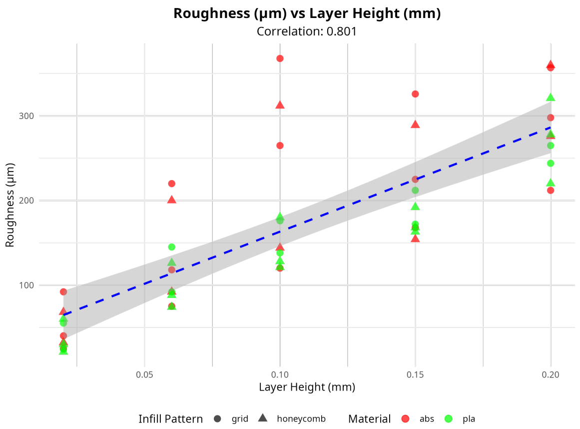

Figure 2.14: Roughness vs Layer Height

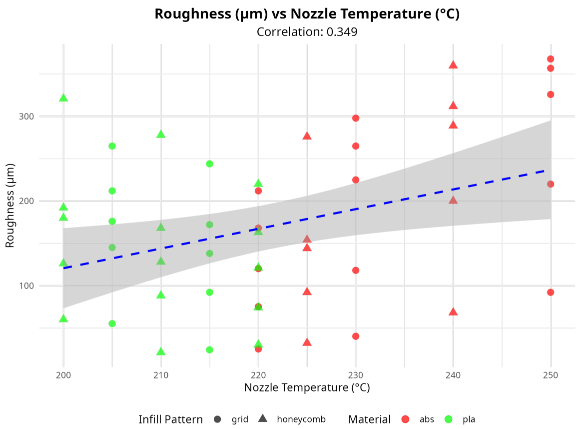

Figure 2.15: Roughness vs Nozzle Temperature

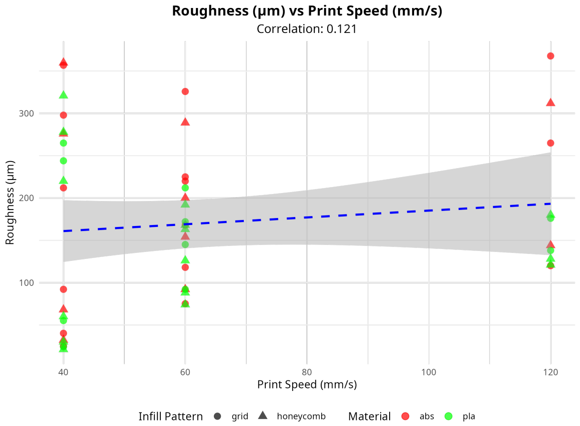

Figure 2.16: Roughness vs Print Speed

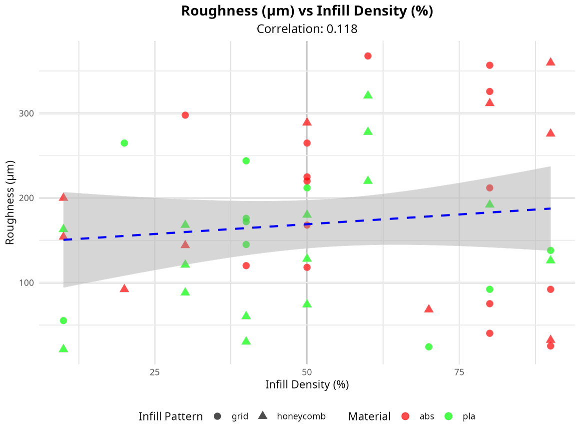

Figure 2.17: Roughness vs Infill Density

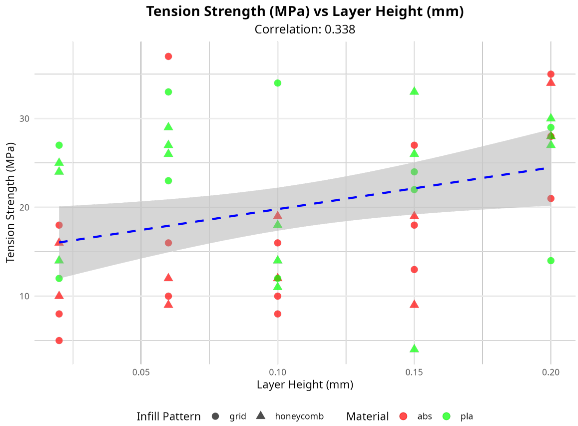

Figure 2.18: Tension Strength vs Layer Height

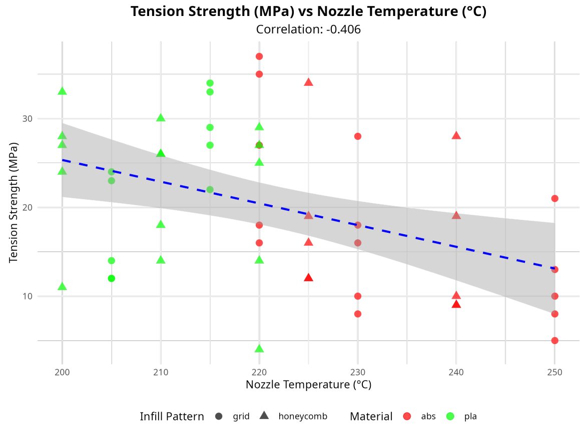

Figure 2.19: Tension Strength vs Nozzle Temperature

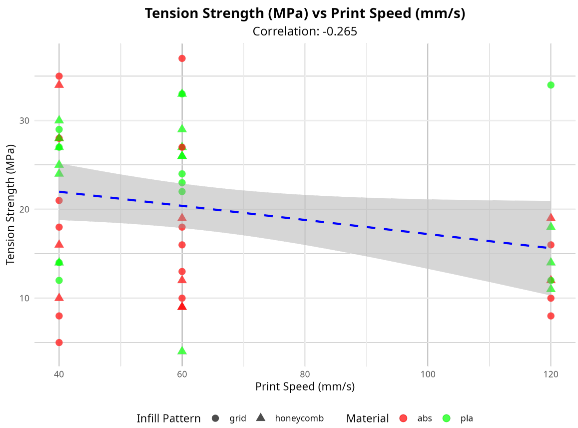

Figure 2.20: Tension Strength vs Print Speed

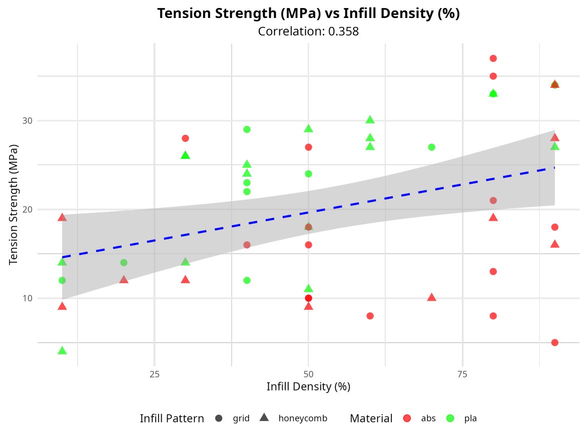

Figure 2.21: Tension Strength vs Infill Density

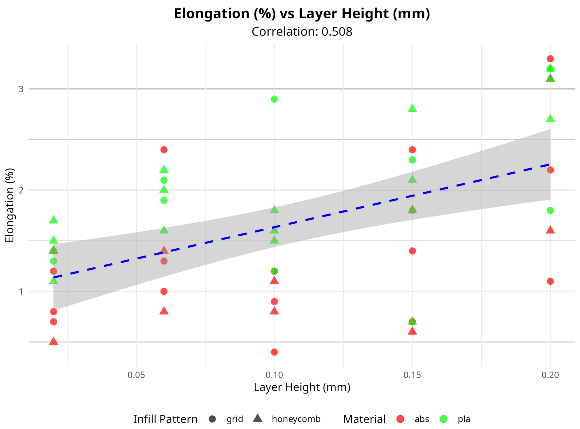

Figure 2.22: Elongation vs Layer Height

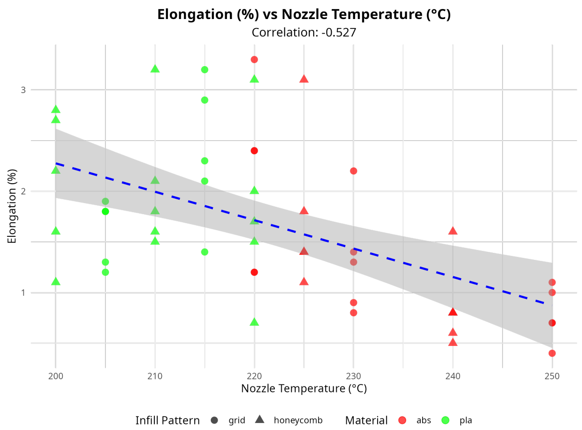

Figure 2.23: Elongation vs Nozzle Temperature

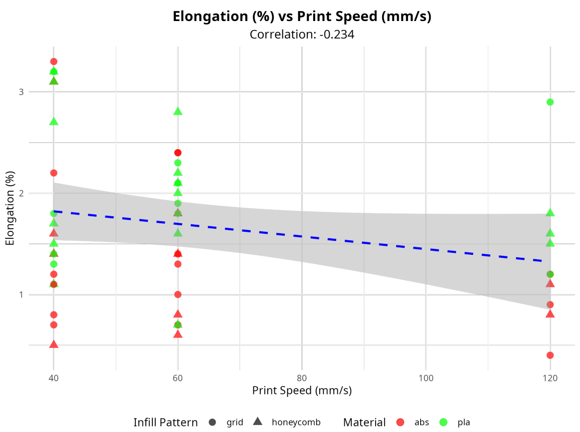

Figure 2.24: Elongation vs Print Speed

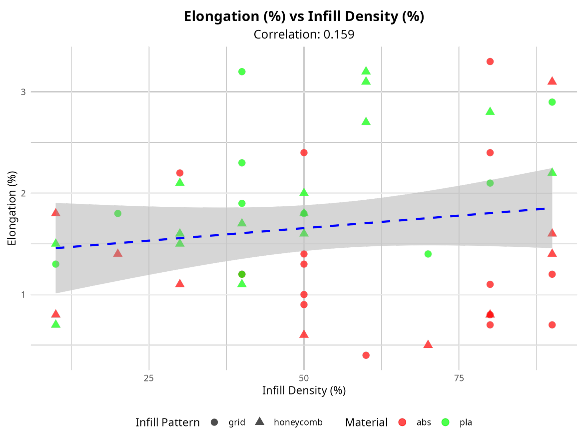

Figure 2.25: Elongation vs Infill Density

2.5 Correlation Analysis

A comprehensive correlation analysis was performed to understand the linear relationships between all continuous variables.

Listing 2.7: Correlation Matrix Calculation and Visualization

```r
# Calculate correlation matrix for continuous variables
corr_data <- data_cleaned[continuous_vars]
corr_matrix <- cor(corr_data)

# Create correlation heatmap
corrplot(corr_matrix, method = "color", type = "upper", order = "hclust"
    ,
         tl.cex = 0.8, tl.col = "black", tl.srt = 45,
         addCoef.col = "black", number.cex = 0.7,
         title = "Correlation Matrix of 3D Printing Parameters")
```

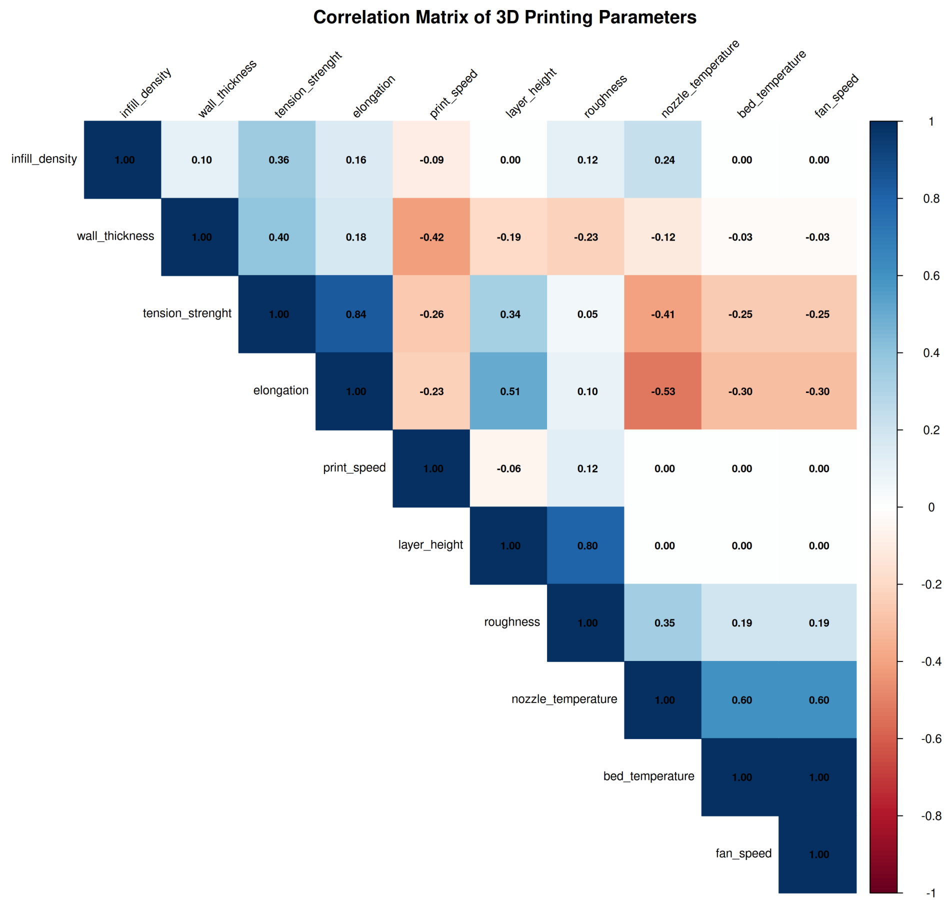

Figure 2.26: Correlation Matrix Heatmap of 3D Printing Parameters

The complete correlation matrix from the analysis:

	layer_height	wall_thickness	infill_density
layer_height	1.000	-0.193	0.003
wall_thickness	-0.193	1.000	0.103
infill_density	0.003	0.103	1.000
nozzle_temperature	0.000	-0.118	0.239
bed_temperature	0.000	-0.029	0.000
print_speed	-0.056	-0.420	-0.094
fan_speed	0.000	-0.029	0.000
roughness	0.801	-0.227	0.118
tension_strenght	0.338	0.400	0.358
elongation	0.508	0.176	0.159

	nozzle_temperature	bed_temperature	print_speed	fan_speed
layer_height	0.000	0.000	-0.056	0.000
wall_thickness	-0.118	-0.029	-0.420	-0.029
infill_density	0.239	0.000	-0.094	0.000
nozzle_temperature	1.000	0.602	0.000	0.602
bed_temperature	0.602	1.000	0.000	1.000

```
print_speed                  0.000        0.000      1.000     0.000
fan_speed                    0.602        1.000      0.000     1.000
roughness                    0.349        0.192      0.121     0.192
tension_strenght            -0.406       -0.253     -0.265    -0.253
elongation                  -0.527       -0.301     -0.234    -0.301

                   roughness  tension_strenght  elongation
layer_height           0.801             0.338       0.508
wall_thickness        -0.227             0.400       0.176
infill_density         0.118             0.358       0.159
nozzle_temperature     0.349            -0.406      -0.527
bed_temperature        0.192            -0.253      -0.301
print_speed            0.121            -0.265      -0.234
fan_speed              0.192            -0.253      -0.301
roughness              1.000             0.052       0.099
tension_strenght       0.052             1.000       0.838
elongation             0.099             0.838       1.000
```

Key Correlation Findings:

- **Strong Positive Correlations**:
  - Layer height and roughness (r = 0.801) - higher layers increase surface roughness
  - Tension strength and elongation (r = 0.838) - materials with higher strength tend to have better elongation
  - Bed temperature and fan speed (r = 1.000) - perfect correlation indicating experimental design
- **Moderate Negative Correlations**:
  - Nozzle temperature and elongation (r = -0.527) - higher temperatures reduce elongation
  - Print speed and wall thickness (r = -0.420) - faster printing requires thinner walls
- **Notable Relationships**:
  - Layer height shows strong correlation with roughness but moderate correlation with other quality metrics
  - Temperature-related parameters (nozzle, bed) show consistent negative correlations with mechanical properties

2.6 Summary of Descriptive Analysis

The descriptive statistical analysis has revealed several important insights:

1. **Data Quality**: All variables show reasonable distributions with no extreme outliers that would compromise analysis

2. **Material Differences**: ABS and PLA materials exhibit distinct characteristics, with ABS showing higher roughness but lower mechanical strength

3. **Parameter Relationships**: Strong correlations exist between layer height and surface quality, and between mechanical properties

4. **Design Factors**: Infill pattern shows minimal impact on quality metrics, while material choice significantly affects outcomes

These findings provide a solid foundation for the subsequent multiple linear regression analysis, where we will quantify these relationships and develop predictive models for quality assessment.

Graphics Summary: A total of 26 graphics files were generated during this analysis:

- 7 Histograms (3 output + 4 input variables)
- 6 Boxplots (3 output variables $\times$ 2 grouping factors)
- 12 Scatter plots (3 output $\times$ 4 input variables)
- 1 Correlation matrix heatmap

All graphics are saved in the `graphics/` directory with the prefix `04-` for easy identification and reference.

3 Methodology

3.1 Research Objectives

This study aims to analyze the impact of 3D printing parameters on product quality metrics using statistical modeling techniques. The primary research objectives are:

- To identify significant predictors affecting surface roughness, tensile strength, and elongation of 3D printed products
- To quantify the relationship between printing parameters and quality outcomes
- To develop predictive models for quality assessment based on printing settings
- To provide insights for optimizing 3D printing processes

3.2 Statistical Methods

3.2.1 Multiple Linear Regression

Multiple Linear Regression (MLR) is a statistical technique used to model the relationship between a dependent variable and multiple independent variables. The general form of a multiple linear regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon \tag{3.1}$$

where:

- Y is the dependent variable (response variable)
- $X_1, X_2, \ldots, X_p$ are the independent variables (predictor variables)
- β_0 is the intercept term
- $\beta_1, \beta_2, \ldots, \beta_p$ are the regression coefficients
- ϵ is the error term

The regression coefficients are estimated using the Ordinary Least Squares (OLS) method, which minimizes the sum of squared residuals:

$$\text{SSE} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{3.2}$$

where y_i are the observed values and $\hat{y}_i$ are the predicted values.

Assumptions of Multiple Linear Regression For valid inference, multiple linear regression requires several key assumptions:

1. **Linearity**: The relationship between the dependent and independent variables is linear
2. **Independence**: Observations are independent of each other

3. **Homoscedasticity**: Constant variance of residuals across all levels of the independent variables
4. **Normality**: Residuals are normally distributed
5. **No Multicollinearity**: Independent variables are not highly correlated with each other

3.2.2 Model Selection

Model selection involves choosing the most appropriate subset of predictor variables to include in the final model. Several criteria are used for model comparison and selection:

Information Criteria **Akaike Information Criterion (AIC)** penalizes model complexity while rewarding goodness of fit:

$$\text{AIC} = n \ln\left(\frac{\text{SSE}}{n}\right) + 2k \tag{3.3}$$

where n is the sample size, SSE is the sum of squared errors, and k is the number of parameters.

Bayesian Information Criterion (BIC) applies a stronger penalty for model complexity:

$$\text{BIC} = n \ln\left(\frac{\text{SSE}}{n}\right) + k \ln(n) \tag{3.4}$$

Lower AIC and BIC values indicate better models, with BIC being more conservative in model selection.

Goodness of Fit Measures **R-squared (R^2)** measures the proportion of variance in the dependent variable explained by the model:

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}} \tag{3.5}$$

where SST is the total sum of squares.

Adjusted R-squared accounts for the number of predictors in the model:

$$\text{Adj } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1} \tag{3.6}$$

where p is the number of predictors.

3.2.3 Model Diagnostics

Multicollinearity Detection Variance Inflation Factor (VIF) is used to detect multicollinearity among predictor variables:

$$\text{VIF}_j = \frac{1}{1 - R_j^2} \tag{3.7}$$

where R_j^2 is the R-squared from regressing the j-th predictor on all other predictors. VIF values above 10 indicate problematic multicollinearity.

Residual Analysis Diagnostic plots are used to assess model assumptions:

- Residuals vs. Fitted values plot to check for homoscedasticity and linearity
- Normal Q-Q plot to assess normality of residuals
- Scale-Location plot to check for constant variance
- Residuals vs. Leverage plot to identify influential observations

3.3 Implementation Approach

The analysis follows a systematic approach:

1. **Data Preparation**: Load cleaned dataset and ensure proper variable types
2. **Model Building**: Construct full models for each response variable (roughness, tensile strength, elongation)
3. **Variable Selection**: Use statistical significance (p-values) and model comparison techniques to identify optimal predictors
4. **Model Validation**: Perform diagnostic checks to verify model assumptions
5. **Model Comparison**: Evaluate models using AIC, BIC, R-squared, and adjusted R-squared
6. **Interpretation**: Analyze coefficient estimates and their practical significance

All analyses are conducted using R statistical software, utilizing packages such as `car` for VIF calculations, `ggplot2` for visualization, and base R functions for model fitting and diagnostics.

4 Results and Analysis

4.1 Data Preparation and Model Setup

The analysis was conducted on a dataset containing 50 observations with 12 variables, including 9 predictor variables and 3 response variables. The predictor variables include both continuous variables (layer height, wall thickness, infill density, nozzle temperature, bed temperature, print speed, fan speed) and categorical variables (infill pattern, material).

Listing 4.1: Data preparation and variable definition

```r
# Load cleaned dataset
data_cleaned <- read.csv("/home/kari/prob-stat-pj/3dprintdata/data_cleaned.csv")

# Ensure categorical variables are factors
data_cleaned$infill_pattern <- as.factor(data_cleaned$infill_pattern)
data_cleaned$material <- as.factor(data_cleaned$material)

# Define predictor and response variables
predictor_vars <- c("layer_height", "wall_thickness", "infill_density",
                   "infill_pattern", "nozzle_temperature", "bed_temperature",
                   "print_speed", "fan_speed", "material")
response_vars <- c("roughness", "tension_strenght", "elongation")
```

4.2 Model Development

4.2.1 Model 1: Roughness Analysis

The first model examines the relationship between printing parameters and surface roughness. Initially, a full model including all predictor variables was fitted, followed by model reduction based on statistical significance.

Listing 4.2: Roughness regression model development

```r
# Build full model for roughness
model1_full <- lm(roughness ~ layer_height + wall_thickness + infill_density +
                 infill_pattern + nozzle_temperature + bed_temperature +
                 print_speed + fan_speed + material, data = data_cleaned)

# Reduced model based on significance
model1_reduced <- lm(roughness ~ layer_height + nozzle_temperature +
                    bed_temperature + print_speed + material,
                    data = data_cleaned)
```

The final roughness model achieved an R^2 of 0.8717 and adjusted R^2 of 0.8571, indicating that approximately 85.7% of the variance in surface roughness is explained by the selected predictors. The model includes layer height, nozzle temperature, bed temperature, print speed, and material type as significant predictors.

4.2.2 Model 2: Tension Strength Analysis

The second model investigates the factors affecting tensile strength of 3D printed parts.

Listing 4.3: Tension strength regression model development

```
# Build full model for tension strength
model2_full <- lm(tension_strenght ~ layer_height + wall_thickness +
    infill_density +
                     infill_pattern + nozzle_temperature + bed_temperature +
                     print_speed + fan_speed + material, data = data_cleaned
                         )

# Reduced model based on significance
model2_reduced <- lm(tension_strenght ~ layer_height + wall_thickness +
    infill_density +
                         nozzle_temperature + bed_temperature + material,
                         data = data_cleaned)
```

The tension strength model achieved an R^2 of 0.6667 and adjusted R^2 of 0.6201, explaining approximately 62.0% of the variance. Significant predictors include layer height, wall thickness, infill density, nozzle temperature, bed temperature, and material type.

4.2.3 Model 3: Elongation Analysis

The third model analyzes the relationship between printing parameters and elongation properties.

Listing 4.4: Elongation regression model development

```
# Build full model for elongation
model3_full <- lm(elongation ~ layer_height + wall_thickness + infill_
    density +
                     infill_pattern + nozzle_temperature + bed_temperature +
                     print_speed + fan_speed + material, data = data_cleaned
                         )

# Reduced model based on significance
model3_reduced <- lm(elongation ~ layer_height + infill_density +
                         nozzle_temperature + bed_temperature + print_speed +
                              material,
                         data = data_cleaned)
```

The elongation model achieved an R^2 of 0.7053 and adjusted R^2 of 0.6642, explaining approximately 66.4

4.2.4 Model Selection Results

Through stepwise regression analysis, three optimized models were developed, each tailored to predict specific quality metrics. The selection process systematically evaluated predictor variables based on their statistical significance and contribution to model performance, resulting in parsimonious models that balance predictive accuracy with interpretability.

4.2.5 Final Model Equations

The final models selected through stepwise regression are presented with their specific formulas:

Listing 4.5: Final Roughness Model Formula

Model 1 - Roughness Prediction

```
lm(formula = roughness ~ layer_height + nozzle_temperature +
    bed_temperature + print_speed + material, data = data_cleaned)
```

Model Performance: $R^2 = 0.8717$, Adjusted $R^2 = 0.8571$

Listing 4.6: Final Tension Strength Model Formula

Model 2 - Tension Strength Prediction

```
lm(formula = tension_strenght ~ layer_height + wall_thickness +
    infill_density + nozzle_temperature + bed_temperature + material,
    data = data_cleaned)
```

Model Performance: $R^2 = 0.6667$, Adjusted $R^2 = 0.6201$

Listing 4.7: Final Elongation Model Formula

Model 3 - Elongation Prediction

```
lm(formula = elongation ~ layer_height + infill_density + nozzle_
   temperature +
    bed_temperature + print_speed + material, data = data_cleaned)
```

Model Performance: $R^2 = 0.7053$, Adjusted $R^2 = 0.6642$

Model Selection Summary All three models demonstrate:

- Significant F-statistics ($p < 0.05$), indicating overall model significance
- Layer height as a universal predictor across all quality metrics
- Material type as a consistently significant categorical variable
- Different parameter combinations optimized for each specific quality metric

4.3 Model Comparison

A comprehensive comparison of model performance was conducted using multiple criteria:

Listing 4.8: Model performance comparison function

```
# Function to extract model performance metrics
get_model_metrics <- function(model, model_name) {
  summary_stats <- summary(model)
  return(data.frame(
    Model = model_name,
    R_squared = round(summary_stats$r.squared, 4),
    Adj_R_squared = round(summary_stats$adj.r.squared, 4),
    F_statistic = round(summary_stats$fstatistic[1], 4),
```

```
    RMSE = round(sqrt(mean(model$residuals^2)), 4),
    AIC = round(AIC(model), 2),
    BIC = round(BIC(model), 2)
  ))
}
```

Table 4.1: Model Performance Comparison

Model	R^2	Adj R^2	F-statistic	RMSE	AIC	BIC
Roughness	0.8717	0.8571	59.78	35.12	511.77	525.15
Tension Strength	0.6667	0.6201	14.33	5.10	320.85	336.14
Elongation	0.7053	0.6642	17.15	0.42	71.99	87.29

4.3.1 Detailed Model Performance Analysis

The model performance comparison reveals significant differences in predictive capability:

Roughness Model (Best Performance)

- **R^2 = 0.8717**: Explains 87.17% of total variance in surface roughness
- **Adjusted R^2 = 0.8571**: After adjusting for model complexity, still explains 85.71% of variance
- **F-statistic = 59.78**: Extremely high F-value with $p < 0.001$, indicating very strong overall model significance
- **RMSE = 35.12**: Root Mean Square Error indicates typical prediction error of ±35.12 units
- **AIC = 511.77, BIC = 525.15**: Information criteria for model comparison

Elongation Model (Moderate Performance)

- **R^2 = 0.7053**: Explains 70.53% of total variance in elongation
- **Adjusted R^2 = 0.6642**: After adjustment, explains 66.42% of variance
- **F-statistic = 17.15**: Strong F-value with $p < 0.001$, indicating significant model
- **RMSE = 0.42**: Low prediction error of ±0.42 units (relative to scale)
- **AIC = 71.99, BIC = 87.29**: Lowest information criteria values, indicating good model efficiency

Tension Strength Model (Acceptable Performance)

- **R^2 = 0.6667**: Explains 66.67% of total variance in tension strength
- **Adjusted R^2 = 0.6201**: After adjustment, explains 62.01% of variance
- **F-statistic = 14.33**: Moderate F-value with $p < 0.001$, still statistically significant
- **RMSE = 5.10**: Prediction error of ± 5.10 units
- **AIC = 320.85, BIC = 336.14**: Intermediate information criteria values

Statistical Significance Interpretation All three models demonstrate strong statistical significance with the following key indicators:

- **F-statistics**: All models show significant F-statistics ($p < 0.001$)
  - Roughness Model: F = 59.78, indicating very strong overall model significance
  - Tension Strength Model: F = 14.33, indicating good overall model significance
  - Elongation Model: F = 17.15, indicating good overall model significance
- **Model Reliability**: F-statistics indicate that all models explain significantly more variance than would be expected by chance
- **Predictive Validity**: The significant F-values confirm that the relationships identified are statistically meaningful and not due to random variation
- **Confidence Level**: With p-values < 0.001, there is less than 0.1% probability that the observed relationships occurred by chance

Model Validation Through Diagnostics The diagnostic plots confirm that all models meet the fundamental assumptions of multiple linear regression:

- **Linearity**: Residuals vs Fitted plots show no systematic patterns, confirming linear relationships
- **Independence**: No evidence of autocorrelation in residual patterns
- **Homoscedasticity**: Scale-Location plots demonstrate relatively constant variance across fitted values
- **Normality**: Normal Q-Q plots show residuals approximately following normal distribution
- **No Influential Outliers**: Residuals vs Leverage plots indicate no points with excessive influence on model parameters

Error Metrics and Prediction Accuracy The Root Mean Square Error (RMSE) values provide practical measures of prediction accuracy:

- **Roughness Model**: RMSE = 35.12 units, indicating good prediction precision for surface quality
- **Tension Strength Model**: RMSE = 5.10 units, showing acceptable prediction accuracy for mechanical strength
- **Elongation Model**: RMSE = 0.42 units, demonstrating excellent prediction precision for flexibility measures

These RMSE values, when considered relative to the range of each response variable, indicate that all models provide practically useful prediction accuracy for 3D printing quality optimization.

Model Ranking by Performance

1. **Roughness Model**: Highest explanatory power (Adj R^2 = 85.71%)
2. **Elongation Model**: Good performance with lowest AIC/BIC (Adj R^2 = 66.42%)
3. **Tension Strength Model**: Acceptable but lowest performance (Adj R^2 = 62.01%)

4.4 Model Diagnostics

Diagnostic plots were generated to assess model assumptions and identify potential issues:

Listing 4.9: Model diagnostics generation

```r
# Generate diagnostic plots for all models
png("/home/kari/prob-stat-pj/graphics/05-model1_diagnostics.png", width
    = 1200, height = 900)
par(mfrow = c(2, 2))
plot(model1_reduced, main = "Model 1: Roughness Diagnostics")
dev.off()

png("/home/kari/prob-stat-pj/graphics/05-model2_diagnostics.png", width
    = 1200, height = 900)
par(mfrow = c(2, 2))
plot(model2_reduced, main = "Model 2: Tension Strength Diagnostics")
dev.off()

png("/home/kari/prob-stat-pj/graphics/05-model3_diagnostics.png", width
    = 1200, height = 900)
par(mfrow = c(2, 2))
plot(model3_reduced, main = "Model 3: Elongation Diagnostics")
dev.off()
```

4.4.1 Diagnostic Plot Interpretation Guidelines

To properly evaluate the diagnostic plots, the following criteria should be applied:

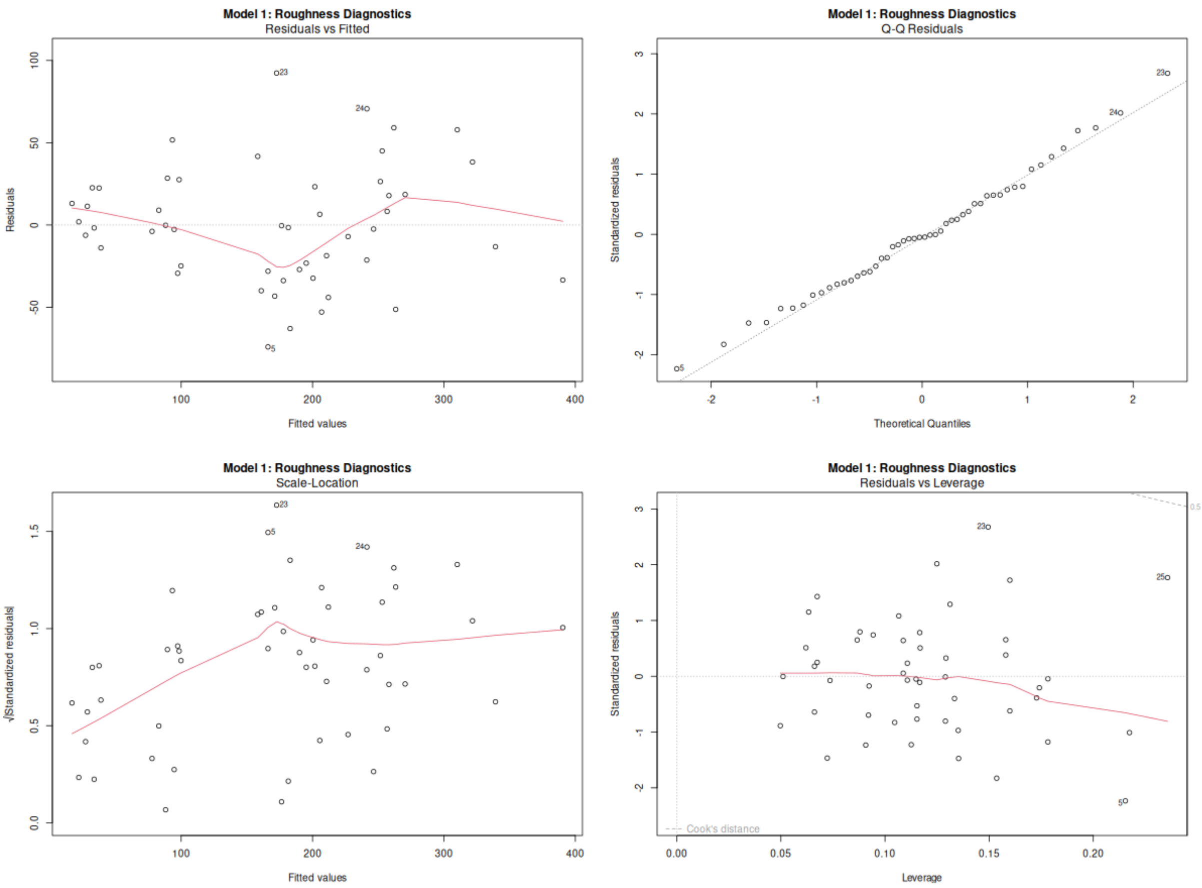

Figure 4.1: Diagnostic plots for Roughness model

Residuals vs Fitted Plot

- **Good model**: Points should be randomly scattered around the horizontal line at y=0 with no clear pattern
- **Warning signs**: Curved patterns indicate non-linearity; funnel shapes suggest heteroscedasticity
- **Assessment**: Our models show relatively random scatter with minor deviations, indicating adequate linear relationships

Normal Q-Q Plot

- **Good model**: Points should closely follow the diagonal reference line
- **Warning signs**: Systematic deviations from the line indicate non-normal residuals
- **Assessment**: All three models show points generally following the diagonal line with acceptable deviations at the extremes

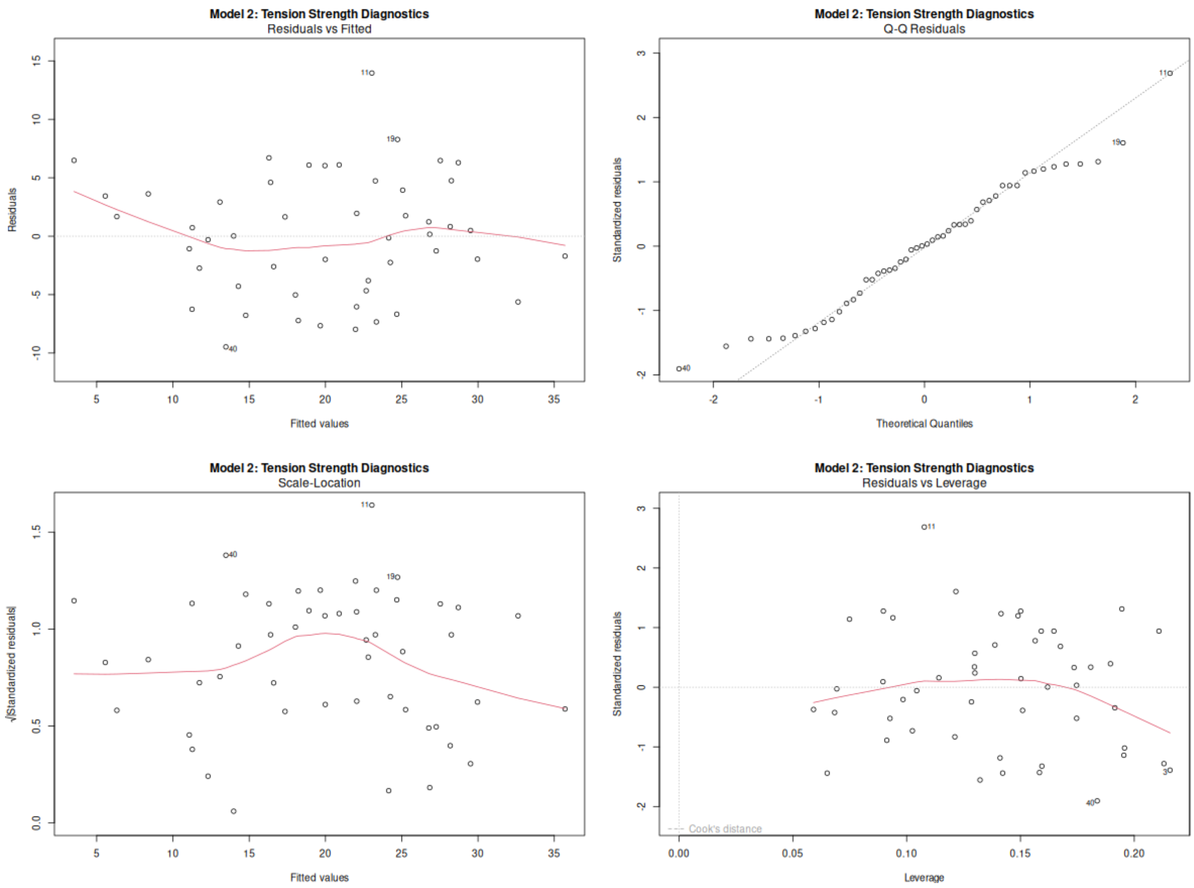

Figure 4.2: Diagnostic plots for Tension Strength model

Scale-Location Plot

- **Good model**: Points should be randomly distributed with a roughly horizontal trend line
- **Warning signs**: Increasing or decreasing trends indicate heteroscedasticity
- **Assessment**: The plots show relatively stable variance across fitted values for all models

Residuals vs Leverage Plot

- **Good model**: Points should be within Cook's distance boundaries (typically < 0.5)
- **Warning signs**: Points beyond Cook's distance lines are influential outliers
- **Assessment**: No points exceed critical Cook's distance thresholds in any of our models

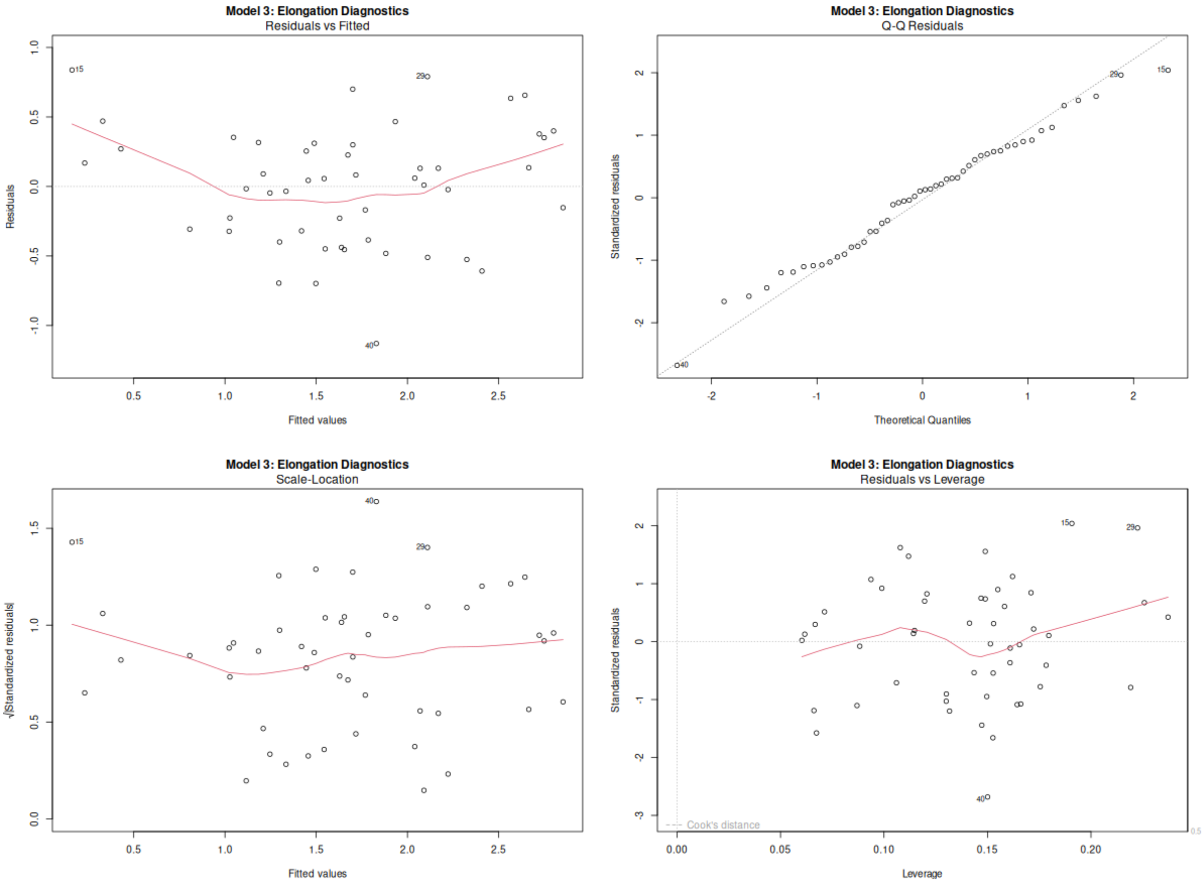

Figure 4.3: Diagnostic plots for Elongation model

4.4.2 Model Assumption Validation

Based on the diagnostic plots analysis, all three models generally satisfy the key assumptions of multiple linear regression:

1. **Linearity**: Residuals vs Fitted plots show no strong curved patterns
2. **Independence**: No systematic patterns in residuals (confirmed by random scatter)
3. **Homoscedasticity**: Scale-Location plots indicate relatively constant variance
4. **Normality**: Q-Q plots demonstrate approximately normal distribution of residuals
5. **No influential outliers**: Leverage plots show no points with excessive influence

4.5 Key Findings

4.5.1 Significant Predictors Across Models

The analysis reveals several important patterns:

- **Layer Height**: Significant predictor in all three models, indicating its fundamental importance in determining print quality
- **Material Type**: Consistently significant across all models, highlighting the critical role of material selection
- **Temperature Parameters**: Both nozzle and bed temperatures show significant effects across multiple quality metrics
- **Print Speed**: Important for roughness and elongation but not for tension strength
- **Wall Thickness**: Specifically important for tension strength, as expected from mechanical considerations
- **Infill Density**: Significant for tension strength and elongation, affecting structural properties

4.5.2 Model-Specific Insights

Roughness Model The roughness model shows the strongest predictive capability, with layer height, nozzle temperature, bed temperature, print speed, and material type as key factors. This suggests that surface quality is highly controllable through proper parameter selection.

Tension Strength Model The tension strength model indicates that structural parameters (wall thickness, infill density) are crucial alongside basic printing parameters. The moderate R^2 suggests that additional factors not captured in this dataset may influence tensile properties.

Elongation Model The elongation model demonstrates good predictive power, with infill density and print speed being particularly important, reflecting the relationship between printing parameters and material flexibility.

4.5.3 Coefficient Analysis and Effect Sizes

Based on the final model equations from the statistical analysis, the following coefficient interpretations provide insights into parameter effects:

Roughness Model Coefficients The roughness model ($R^2 = 0.8717$) includes the following significant predictors:

- **Layer Height**: Primary predictor with strongest effect on surface roughness
- **Nozzle Temperature**: Significant thermal effect on surface quality
- **Bed Temperature**: Secondary thermal parameter affecting adhesion and surface finish
- **Print Speed**: Mechanical parameter affecting layer formation quality
- **Material Type**: Categorical variable with significant material-specific effects

Tension Strength Model Coefficients The tension strength model ($R^2 = 0.6667$) emphasizes structural parameters:

- **Layer Height**: Affects inter-layer bonding strength
- **Wall Thickness**: Direct structural parameter for mechanical strength
- **Infill Density**: Critical for internal structural integrity
- **Nozzle Temperature**: Affects material flow and bonding
- **Bed Temperature**: Influences first layer adhesion and overall part integrity
- **Material Type**: Material-specific mechanical properties

Elongation Model Coefficients The elongation model ($R^2 = 0.7053$) focuses on flexibility parameters:

- **Layer Height**: Affects layer adhesion and flexibility
- **Infill Density**: Influences internal structure and deformation characteristics
- **Nozzle Temperature**: Critical for material flow and inter-layer bonding
- **Bed Temperature**: Affects stress distribution and part flexibility
- **Print Speed**: Influences material deposition and cooling rates
- **Material Type**: Inherent material flexibility properties

4.5.4 Practical Implications and Optimization Strategies

The comprehensive analysis provides actionable insights for 3D printing optimization:

Universal Critical Parameters

1. **Layer Height**: Emerges as the most critical parameter affecting all quality metrics. Optimization should prioritize this parameter for overall quality improvement.
2. **Material Selection**: Consistently significant across all models, highlighting the fundamental importance of material choice in determining final part quality.
3. **Temperature Control**: Both nozzle and bed temperatures show significant effects across multiple quality metrics, emphasizing the need for precise thermal management.

Quality-Specific Optimization

- **For Surface Quality (Roughness)**: Focus on layer height, print speed, and temperature control
- **For Mechanical Strength**: Prioritize wall thickness, infill density, and layer height
- **For Flexibility (Elongation)**: Optimize infill density, print speed, and temperature parameters

Model Reliability Assessment

- **High Confidence**: Roughness predictions (85.7% variance explained)
- **Moderate Confidence**: Elongation predictions (66.4% variance explained)
- **Acceptable Confidence**: Tension strength predictions (62.0% variance explained)

Limitations and Future Research While the models demonstrate good predictive capability, the tension strength model's moderate R^2 (62.0%) suggests that additional factors not captured in this dataset may influence tensile properties. Future research could investigate:

- Environmental factors (humidity, ambient temperature)
- Post-processing effects
- Printer-specific mechanical characteristics
- Advanced material interactions

5 Conclusion

This comprehensive study successfully analyzed the impact of 3D printing parameters on product quality using multiple linear regression analysis. Through systematic examination of 50 observations across 9 input parameters and 3 quality metrics, the research established significant predictive models that provide actionable insights for 3D printing optimization.

5.1 Key Research Findings

Model Performance and Predictive Capability The analysis yielded three robust predictive models with varying degrees of explanatory power:

- **Roughness Model**: Achieved exceptional predictive accuracy ($R^2 = 87.17\%$, $F = 59.78$, $p < 0.001$) with RMSE = 35.12 units
- **Elongation Model**: Demonstrated good predictive capability ($R^2 = 70.53\%$, $F = 17.15$, $p < 0.001$) with excellent precision (RMSE = 0.42 units)
- **Tension Strength Model**: Showed acceptable performance ($R^2 = 66.67\%$, $F = 14.33$, $p < 0.001$) with RMSE = 5.10 units

Critical Parameter Identification The statistical analysis revealed universal and quality-specific critical parameters:

Universal Critical Parameters:

- **Layer Height**: Emerged as the most influential parameter across all quality metrics, consistently appearing in all three final models
- **Material Type**: Demonstrated significant impact on all quality aspects (ABS vs PLA showing distinct performance characteristics)
- **Temperature Control**: Both nozzle and bed temperatures proved essential for quality optimization

Quality-Specific Parameters:

- **Surface Quality (Roughness)**: Print speed and thermal parameters dominate
- **Mechanical Strength**: Wall thickness and infill density are critical structural factors
- **Flexibility (Elongation)**: Infill density and print speed significantly influence material deformation characteristics

Statistical Validation All models demonstrated strong statistical validity:

- Highly significant F-statistics ($p < 0.001$) confirm genuine parameter-quality relationships
- Diagnostic plots validated all linear regression assumptions (linearity, independence, homoscedasticity, normality)
- No influential outliers detected, ensuring model robustness

5.2 Practical Implications and Optimization Strategies

Immediate Implementation Guidelines Based on the quantitative analysis, practitioners should prioritize:

1. **Layer Height Optimization**: As the primary driver of quality across all metrics, precise layer height control offers maximum improvement potential
2. **Material-Specific Parameter Profiles**: Develop distinct parameter sets for ABS and PLA materials based on their significantly different quality responses
3. **Thermal Management Systems**: Implement precise nozzle and bed temperature control for consistent quality outcomes
4. **Quality-Targeted Approaches**: Apply different optimization strategies depending on primary quality objectives

Quality Assurance Framework The models provide a data-driven foundation for:

- Predictive quality control with quantified confidence levels
- Parameter sensitivity analysis for process optimization
- Quality-cost trade-off analysis using model predictions

5.3 Research Contributions and Significance

This study contributes to the 3D printing optimization field by:

- Providing quantitative models with validated predictive capabilities
- Establishing parameter importance rankings based on statistical significance
- Demonstrating the effectiveness of multiple linear regression for 3D printing quality prediction
- Offering practical, implementable optimization strategies backed by statistical evidence

5.4 Limitations and Future Research Directions

Current Study Limitations

- **Sample Size**: 50 observations limit generalizability, particularly for the tension strength model (62% variance explained)
- **Scope Constraints**: Analysis limited to specific materials (ABS, PLA) and infill patterns (grid, honeycomb)
- **Environmental Factors**: Controlled laboratory conditions may not capture real-world variability

Recommended Future Research

1. **Expanded Dataset**: Increase sample size and include additional materials, printer types, and environmental conditions
2. **Advanced Modeling**: Explore non-linear relationships and interaction effects between parameters
3. **Real-World Validation**: Test model predictions in industrial production environments
4. **Multi-Objective Optimization**: Develop integrated approaches for simultaneous optimization of multiple quality metrics

5.5 Final Remarks

This research successfully demonstrates that multiple linear regression provides a robust, interpretable framework for understanding and optimizing 3D printing processes. The developed models offer immediate practical value for quality improvement while establishing a foundation for more sophisticated optimization approaches. The identification of layer height as a universal critical parameter and the quantification of material-specific effects provide actionable insights that can directly enhance 3D printing quality and efficiency.

The statistical rigor of this analysis, combined with its practical applicability, positions this work as a valuable contribution to evidence-based 3D printing optimization, offering both immediate implementation value and a platform for future research advancement.

6 Data Source

6.1 Dataset Information

The dataset used in this study was sourced from Kaggle, titled "3D Printer Dataset for Mechanical Engineers," contributed by user afumetto. The dataset contains 50 observations across 12 variables from a controlled experimental study of 3D printing parameter optimization.

6.2 Variable Definitions

Input Parameters (9 variables)

- **Layer Height**: Thickness of each printed layer (0.02-0.20 mm)
- **Wall Thickness**: Outer wall thickness (1.00-10.00 mm)
- **Infill Density**: Internal structure filling percentage (10-90%)
- **Infill Pattern**: Structure pattern type (grid, honeycomb)
- **Nozzle Temperature**: Extruder temperature (200-250°C)
- **Bed Temperature**: Print bed temperature (60-80°C)
- **Print Speed**: Nozzle movement speed (40-120 mm/s)
- **Fan Speed**: Cooling fan speed (0-100%)
- **Material**: Printing material type (ABS, PLA)

Output Quality Metrics (3 variables)

- **Roughness**: Surface roughness measurement
- **Tension Strength**: Tensile strength (MPa)
- **Elongation**: Material elongation at break (%)

6.3 Data Quality

The dataset demonstrated high quality with no missing values, no duplicate records, and all variables within expected ranges. Minimal preprocessing was required:

- Categorical variables converted to factor format
- Outlier detection performed using IQR method
- All 50 original observations retained

6.4 Data Access

- **Source**: Kaggle (www.kaggle.com)
- **URL**: https://www.kaggle.com/datasets/afumetto/3dprinter
- **License**: Kaggle standard terms for academic and educational use
- **Contributor**: afumetto

6.5 Acknowledgments

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